

6. Applications of Integrals

6.4. Work

6.4 b Learning Objectives:

- I can understand the definitions of work and force.
- I understand how to use integration to describe work.
- I can describe the work done when pumping fluids out of containers of varying shapes.

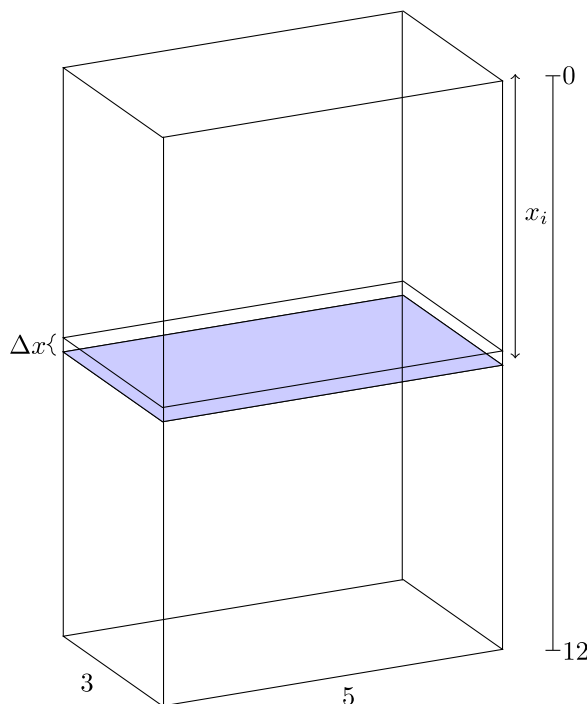
Another physical system we can calculate the work done in is when pumping liquids out of a tank.

6.4.5. Pumping Liquids

Example 6.4.1

A rectangular tank full of water has height 12 ft, width 3 ft, and length 5 ft. Find the work done in pumping all of the water to the top of the tank given that the density of water is 62.4 lbs ft^{-3} .

 **Solution**



Consider pumping a single layer of water out at a time. The work done to pump the i th layer of water out is $W_i = F_i d_i$. We will say the i th layer has to be pumped x_i ft out of the tank, and x_i goes from 0 to 12.

Then the force on each layer is the weight of the slice which can be found by multiplying density and volume: $F_i = 62.4 \text{ lbs ft}^{-3} V_i$. Now $V_i = (3)(5)(\Delta x)$, where Δx is the height of the slice. Altogether $F_i = 62.4(15)\Delta x$ lb.

Then the work on each slice is $W_i = 62.4(15)\Delta x x_i$ lbft.

The work is then the limit of the sums of the layers


$$\begin{aligned} W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (62.4)(15) \Delta x x_i \\ &= \int_0^{12} (15)(62.4)x \, dx \end{aligned}$$

6.4 b Section Summary: